

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	XXXXXX
Project Title	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution introduces an AI-powered Voice-Based Concept Understanding Analyzer that evaluates a learner's conceptual understanding using speech processing and semantic analysis. The system converts spoken explanations into text, compares them with reference concepts, analyzes voice features, generates performance scores, and provides downloadable PDF reports.

Project Overview

Topic	Description
Objective	Develop an intelligent application that evaluates conceptual understanding using voice input and AI-based semantic analysis.
Scope	The system supports voice upload, transcription, semantic similarity analysis, audio feature extraction, score calculation, report generation and visualization.

Problem Statement

Topic	Description
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Description	Traditional assessments often measure memorization rather than conceptual understanding and require significant manual effort.
Impact	An automated voice-based evaluation system improves assessment quality, reduces manual work and delivers instant feedback.

Proposed Solution

Topic	Details
Approach	Use Speech-to-Text, Sentence Transformers, audio feature extraction and scoring algorithms to evaluate conceptual understanding.
Key Features	- Voice recording upload - Speech-to-Text transcription - Semantic similarity analysis - Audio feature extraction - Performance scoring - Downloadable PDF reports - Interactive dashboard

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Processor	CPU	Intel Core i5 / Ryzen 5 or higher
Memory	RAM	8 GB or above
Storage	Disk Space	10 GB free SSD space
Software		

Framework	Application Framework	Streamlit
Libraries	Python Libraries	OpenAI Whisper, Sentence Transformers, Librosa, NumPy, Pandas, Matplotlib, FPDF
Development Environment	IDE	Visual Studio Code
Data		
Dataset	Reference Concepts & Audio Samples	JSON files, WAV/MP3 recordings